

## **Practitioner's Commentary: Salt Storage Done Improperly, Safety Panel Says**

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### **Piles are Too Large, Could Shift Dangerously, State Told**

A violation cited by the Indiana Occupational Safety and Health Administration will result in a change in state highway policy and may force a \$600,000 purchase of conveyors for loading salt.

"We're telling our people—don't climb a mountain to put the salt in," said Donald W. Lucas, deputy director of operations for the Indiana Department of Highways.

The storage practice in Indiana's 80 salt domes has been the same since the early 1970s.

Employees drive on ramps of salt leading into the domes and pile salt up to 25 or 30 feet with buckets attached to 10- or 15-ton front-end loaders.

"Even though we haven't had an accident history, it's probably not a good idea to use front-end loaders to pile salt any higher than 15 feet," Lucas explained.

Several employees, whose names are not part of the IOSHA record, made informal complaints about the practice concerning one of the 50-foot-high domes in the Crawfordsville District.

"In piling or storing salt in this manner, if the end loader gets too close to the edge of the pile, the salt can shift," says the IOSHA report, which followed two inspections in July and August.

The shifting pile of salt could throw the loader against retaining walls that reinforce the dome.

There were at least two incidents in Crawfordsville involving front-end loaders slipping on salt piles, but neither resulted in injury to the operator nor damage to equipment, said Larry R. Vaughan, district maintenance engineer.

Mechanical conveyors, similar to ones used for fertilizer and grain, may be a partial answer. The conveyors, however, cost about \$30,000 apiece.

And because salt is so corrosive, a stainless steel model may last only one season.

The department is discussing the purchase of up to 20 conveyors, for \$600,000, that could be transported by truck to several districts and shared.

IOSHA's citation came as a surprise to highway officials and one New Jersey manufacturer. According to state law, an employer must provide a place to work free of hazards likely to cause injury.

"I don't think the citation is worthy of the fuss," said David M. Wilcox, district engineer in Crawfordsville.

"Everybody agrees. Some sort of mechanical filling process would be great. Why change it for the sake of spending money. Every dollar spent needlessly takes it away from road service. But I'm not in a position to argue."

Only one of the district offices for the department has been cited, even though the same loading method is practice around the state. But as one official noted, "If it's a problem for Crawfordsville, it's a problem for everybody."

Lucas said conveyor were purchased for another district about two years ago when highway officials were attempting to address the safety factor without any mandates from IOSHA.

By citing budget constraints, the highway department was able to have the violation abated until May 26, 1987. The domes have already been loaded with 240,000 tons of salt for this winter season.

Dome Corporation of America, a New Jersey company, is just one of the suppliers for Indiana. Dome Corp. has sold 2,000 salt domes nationwide and only 1% have conveyor systems for loading.

"Some of them have gone with conveyors. It's recent. And it's an expensive proposition. You're dealing with a corrosive material," said Stan Milic, president of Dome Corp.

"People have not been hurt. Common sense will tell you—you can only climb so high," he added. Indiana is the only state Milic is aware of that has been cited for a safety problem.

Robert L. McCreary, commissioner of the Indiana Labor Department, said the highway department has been "agreeable" to making the needed changes in storage policies.

"I don't think there will be a problem. They're trying to get this resolved. It's a lot of money, but I'm convinced we'll work it out," said McCreary.